GroundFall-US: An Open Dataset of Ground-Failure Incidents at U.S. Mines, 2000–2026, with Rates and a Failure-Mode Coding Layer

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Abstract

Falls of roof, rib, face, highwall, and slope remain among the most lethal hazards in U.S. mining, yet the federal record of these events is spread across separate files that are hard to join and do not record how the ground failed. We present GroundFall-US, an open, documented dataset that joins the Mine Safety and Health Administration (MSHA) Accidents, Mines, and yearly Employment/Production files into 29,851 ground-failure records from 2000 to 2026, with injury and fatality rates per hour worked by sector, mine type, commodity, and state, and a rule-based failure-mode layer (rockfall, raveling, planar, wedge, toppling, rotational slide; blasting, water, and scaling triggers) for surface metal and nonmetal narratives. Ground failures account for 4.1% of reportable injuries but 13.3% of deaths in MSHA's accident file. They are rare at surface mines but lethal there: 14.9% of surface ground-failure injuries were fatal, against 1.3% in underground coal. Injury rates fell in every setting, but the fatality rate at underground metal and nonmetal mines did not fall with them. At surface metal and nonmetal mines, 17% of narratives name no mechanism, only 7% name a structurally controlled one, and MSHA's own classification misses comparable events: screening other categories yields 226 candidate ground failures, 2.5 times the number selected. We release the data, code, rules, and source fingerprints under open licenses, and recommend structured ground-failure fields in federal accident reporting as the basis for early warning under 30 CFR 56.3130.

Introduction

When rock or soil gives way at a mine, the people nearest the face, wall, or bench have seconds or less to react. U.S. regulations address this hazard directly: at surface metal and nonmetal mines, 30 CFR 56.3130 requires that mining methods maintain the stability of walls, banks, and slopes where persons work or travel (Code of Federal Regulations, 2026b), and companion standards cover scaling, inspection, and ground support. Every operator must report accidents, injuries, and certain non-injury events, including unplanned roof falls, to MSHA under 30 CFR Part 50 (Code of Federal Regulations, 2026a). MSHA publishes those reports as open data, refreshed weekly (Mine Safety and Health Administration, 2026).

The record exists, but it is not easy to use for ground control. The accident reports, the list of mines, and the hours each mine worked live in three separate files with different keys and column names. The accident file classifies each event by the circumstance that most directly contributed, so "fall of face, rib, pillar, side, highwall" and "fall of roof or back" can be counted, but it does not record whether a highwall failed as a wedge, a toppling column, or a slump, or what set it off. That information exists only in a free-text narrative of at most 384 characters. Studies of ground falls have therefore centred on underground coal, where roof and rib falls dominate and NIOSH has tracked trends for decades (Mark, Pappas, and Barczak, 2009; Pappas and Mark, 2012; Khademian et al., 2022). Surface walls and slopes, the subject of 56.3130, have received less systematic national attention.

We ask: ***Where, at which kinds of U.S. mines, and how do ground failures that injure miners occur, and what does the federal record allow us to say about the failure mechanism?*** We answer with GroundFall-US, an open dataset built entirely from public MSHA files, and report three findings:

1. **Ground failures are a small share of injuries but a large share of deaths, and surface failures are the deadliest.** They are 4.1% of reportable injuries but 13.3% of fatalities in MSHA's accident file for 2000–2026 (21.9% in coal). A ground-failure injury is 3.2 times as likely to be fatal as the average injury, and at surface mines 14.9% are fatal.

2. **Injury rates have fallen everywhere, but not evenly, and not for deaths.** The underground coal injury rate fell 37% between 2000–04 and 2020–25. At underground metal and nonmetal mines the injury rate fell 59%, while the fatality rate stayed at 6.0 and 5.5 deaths per 100 million hours in the first and last periods. Injuries concentrate in a few states: WV, KY, IL account for 70% of underground coal ground-failure injuries.

3. **The federal record rarely says how surface walls fail, and sometimes does not say that they failed.** Of 92 surface metal and nonmetal narratives, rules flag 41 as rock-

fall and 21 as wall or bank collapse with no mechanism stated; only 6 name a structurally controlled mechanism, and 21% name any trigger. A screen of other MSHA categories finds 226 further narratives that read as ground failures, including 13 fatal records, filed as powered haulage, machinery, and other circumstances.

Contributions. (i) An open, versioned dataset joining three MSHA files with documented keys, provenance, and quality checks; (ii) a rate table that divides counts by hours worked for every year, sector, mine type, commodity, and state; (iii) a transparent, editable rule set that labels surface narratives by failure mode and trigger and records the phrase behind every label; (iv) a candidate list of likely ground failures outside MSHA's fall categories, for review; and (v) recommendations for structured reporting.

Related work. NIOSH researchers have documented the decline of underground coal ground-fall fatalities and the persistence of rock falls between bolts and rib falls (Mark, Pappas, and Barczak, 2009), profiled roof and rib fall trends over a decade, showing roof-fall injury rates falling by about half while rib-fall rates held steady (Pappas and Mark, 2012), and analysed 8,303 ground-fall narratives from 2010 to 2019 across coal and metal/nonmetal mines in five categories, including highwall failure (Khademian et al., 2022). Broader work uses MSHA data to analyse equipment-related injuries (Groves, Kecojevic, and Komljenovic, 2007), to weigh severity measures such as lost workdays (Coleman and Kerkering, 2007), and to classify narratives with machine learning (Ganguli, Miller, and Pothina, 2021). Rock slope engineering has a mature vocabulary for failure mechanisms: planar, wedge, toppling, and circular failure (Wyllie and Mah, 2004; Goodman and Bray, 1976; Read and Stacey, 2009), within landslide classifications that distinguish falls, topples, slides, and flows (Varnes, 1978; Hungr, Leroueil, and Picarelli, 2014). GroundFall-US connects the two: it attaches that vocabulary to the national injury record, and releases the result as reusable data following FAIR principles (Wilkinson et al., 2016) with a datasheet-style codebook (Gebru et al., 2021).

Data and Methods

We built GroundFall-US as a reproducible pipeline of numbered scripts. Each step writes its outputs and a quality report; every number in this paper is generated by the pipeline into a statistics file that the manuscript reads, so the text cannot drift from the data.

Sources

MSHA open data. We used three files from MSHA's Open Government Data portal (Mine Safety and Health Administration, 2026), all in the 2026-10-09 weekly edition and all U.S. government works in the public domain (Table 1). The Accidents file holds every Part 50 report: 275,220 records dated 2000 through 09/23/2026. The Mines file lists every mine MSHA has assigned an identifier, with location, current status, and primary commodity. The yearly Employment/Production file gives hours worked by mine-operator employees for each mine, subunit (underground, surface, mill, and so on), and year. We record the size and SHA-256 fingerprint of each file (Appendix F) and archive the files with the release, because MSHA replaces them weekly and may revise past records.

File	Content and role
Accidents	One row per Part 50 report: mine, date, subunit, MSHA classification, degree of injury, contractor, narrative ($\leq$384 characters). Numerator and text.
Mines	One row per mine: state, county, coal or metal/nonmetal, mine type, primary commodity (canvass) and SIC. Grouping attributes.
Employment/ Production, yearly	One row per mine, subunit, and year: hours worked and average employees. Denominator.

Table 1: The three MSHA source files (2026-10-09 edition).

Selection and joins

Ground failure. A record enters GroundFall-US when MSHA classified the circumstance as "fall of roof or back" or "fall of face/rib/pillar/side/highwall". This yields 29,851 records: 27,472 roof or back falls and 2,379 face, rib, pillar, side, or highwall falls. We read every record of the neighbouring category "falling/sliding/rolling materials" in an early test and found it describes rocks off screens, tools, and equipment rather than ground, so it is excluded. Selection on MSHA's classification is reproducible and auditable, but it inherits MSHA's coding decisions; we measure what it misses in Finding 3.

Joins. Each record joins to the Mines file on the seven-digit mine identifier (100.0% matched) and to hours worked on mine, subunit, and calendar year (99.65% matched). Mine type follows the accident subunit: underground (01); surface (02, 03, 04, 05, 06, 12); mill or plant (30); shop or yard (17); office (99). Commodity follows MSHA's primary canvass code: coal, metal, nonmetal, stone, or sand and gravel. State, commodity, and name are the mine's current values. The released column names follow MSHA's definition files; the files MSHA serves use some different names, which the reader maps.

Severity. MSHA's degree-of-injury code separates fatalities (01), lost-time injuries (02–04), other reportable injuries (05–06), and accidents without injury (00). Part 50 requires some events to be reported even when nobody is hurt, notably unplanned roof falls at or above the anchorage zone underground. These no-injury reports are 68.1% of all ground-failure records, and almost all are underground. We therefore keep them in the incident table but report *injury* rates (degrees 01–06) as the headline measure, so that differences

in reporting obligations do not masquerade as differences in hazard.

Rates. The rate table divides counts by hours worked for every combination of year, sector, mine type, commodity, and state. We report the injury rate per 200,000 hours (MSHA's convention, roughly 100 full-time workers for a year) and the fatality rate per 100 million hours. Contractor accidents (719 records) stay in the incident table but are excluded from rate numerators, because the hours file covers mine-operator employees only. The partial year 2026 is excluded from rates. For period fatality rates, where counts are small, we give exact Poisson 95% intervals.

Quality checks. The pipeline writes a QA report with row counts at each stage, join match rates with unmatched identifiers listed, range checks, and named records for hand verification. As an external check we compared our 2010–2019 subset with the most comparable published count we found (Khademian et al., 2022): we find 8,572 ground-failure records, 5,188 without injury, and 50 fatalities, against their 8,303, 5,175, and 46. Total records differ by 3.2% and no-injury records by 0.3%, consistent with MSHA's revisions to past records and small differences in selection.

Failure-mode coding layer

Scope. The coding layer covers ground-failure records at surface metal and nonmetal operations (subunits 02, 03, 06, 12): 92 narratives. This is the population regulated by 56.3130 and the one for which the failure mechanism matters most for slope design and monitoring.

Taxonomy. Labels follow rock slope engineering usage (Wyllie and Mah, 2004; Read and Stacey, 2009; Goodman and Bray, 1976; Hungr, Leroueil, and Picarelli, 2014): *rockfall* (individual rocks or blocks falling or rolling from a face, bench, or wall); *raveling* (progressive loss of small or loose material); *planar* sliding on a single discontinuity; *wedge* sliding on two intersecting discontinuities; *toppling* (forward rotation of columns or blocks); *rotational slide* (slumping on a curved surface, with tension cracks or toe heave). Reading real narratives showed that many describe a wall or bank that "collapsed" or "gave way" without any mechanism, so we added *collapse, mechanism unstated* rather than force those records into a mechanism the text does not support. Records that match no pattern are *unspecified*. Triggers are multi-label: *blasting*, *water* (rain, saturation, freeze–thaw), and *scaling* (also taken from MSHA's activity field).

Rules. Each label is a set of case-insensitive regular expressions with optional exclusions, 46 patterns in all (Appendix B), held in an editable configuration file rather than in code. Modes are tested in a fixed precedence, from structurally specific (wedge, planar, toppling, rotational slide) through raveling and rockfall to collapse; the first match wins. For every label the dataset stores the exact phrase that fired it, so any user can see why a record was labelled and disagree with it. The first author wrote and approved the rules (version 1.0); an initial draft was prepared with AI assistance and revised against real narratives.

Setting	Records	No inj.	Inj.	Deaths	Fatal %
Coal, underground	28,034	19,643	8,343	109	1.3
Coal, surface	161	78	82	14	17.1
M/NM, underground	1,561	594	958	29	3.0
M/NM, surface	92	12	79	10	12.7

Table 2: Ground-failure records by setting and severity, 2000–2026, contractors included. "No inj." are reportable falls without injury; "Inj." are injuries, MSHA degrees 01–06; "Fatal %" is deaths as a share of injuries.

Supplement screen. To measure what MSHA's classification misses, we applied the same mode patterns to every surface metal and nonmetal narrative in all other MSHA categories, kept matches that mention a ground element (highwall, wall, bank, slope, bench, face, berm, ledge), and removed person falls and plant contexts (crusher, screen, conveyor, hopper). The resulting candidates are listed for expert review; none enters the v1.0 counts.

Definitions. *Injury rate* is degree 01–06 records per 200,000 hours; *fatality rate* is degree 01 records per 100 million hours; the *case-fatality share* is fatalities divided by injuries (degrees 01–06), in percent. *M/NM* is metal and nonmetal. Periods are 2000–04, 2005–09, 2010–14, 2015–19, and 2020–25. All counts include contractors unless a rate is quoted.

Limitations. The dataset sees only what operators reported; a state study comparing workers' compensation claims with Part 50 found that the program captured about 43% of acute injuries (Friedman et al., 2015). No-injury reporting obligations differ between settings. Mine attributes are current rather than historical. Surface metal and nonmetal counts are small, so yearly rates swing on single events. Narratives are short, written by operators, and describe what happened to a person rather than how the ground failed. The coding rules were reviewed by their author, but v1.0 does not report a formal agreement statistic against blind hand labels; Finding 3 reports the errors we found by inspection instead.

Finding 1: Ground Failures Are Few but Deadly, and Surface Failures Are the Deadliest

Of 228,860 reportable injuries in MSHA's accident file from 2000 to 2026, 4.1% were ground failures. Of 1,215 deaths, 13.3% were. In coal the share of deaths is 21.9% (of 562); in metal and nonmetal mining it is 6.0% (of 653). Across all settings, 1.71% of ground-failure injuries were fatal, against 0.53% for all injuries, a ratio of 3.2.

Underground coal dominates the count. Underground coal produced 28,034 of the 29,851 records and 8,343 injuries (Table 2). Most of those records, 70.1%, are reportable roof falls in which nobody was hurt. This reflects Part 50's

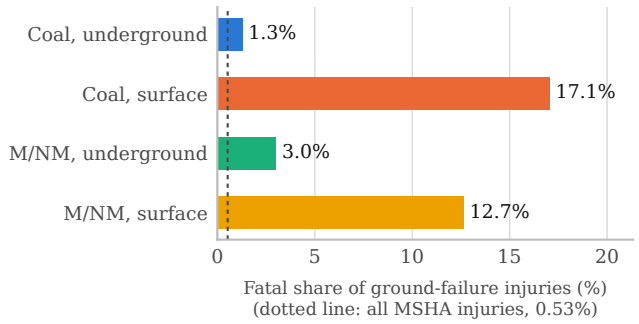


Figure 1: Case-fatality share of ground-failure injuries by setting, 2000–2026. Surface ground-failure injuries are 10 times as likely to be fatal as underground ones (14.9% against 1.5%).

	00–04	05–09	10–14	15–19	20–25
Injuries per 200,000 hours					
Coal, underground	0.857	1.197	0.766	0.650	0.541
Coal, surface	0.0140	0.0157	0.0039	0.0034	0.0038
M/NM, underground	0.412	0.517	0.355	0.206	0.170
M/NM, surface	0.0046	0.0041	0.0031	0.0033	0.0012
Coal, underground	7.0	8.8	4.2	3.6	3.1
Coal, surface	0.6	1.0	0.6	0.8	0.5
M/NM, underground	6.0	6.3	4.5	2.8	5.5
M/NM, surface	0.4	0.1	0.3	0.1	0.1

Table 3: Ground-failure injury rates (upper block) and fatality rates per 100 million hours (lower block) by five-year period, mine-operator employees.

reporting obligation for unplanned roof falls, not only hazard; at surface metal and nonmetal mines, by contrast, only 13.0% of records are injury-free.

Surface failures are rare and lethal. Surface coal and surface metal and nonmetal mines together produced only 161 ground-failure injuries in 27 years, but 24 of them were deaths: 14.9% (Figure 1). The case-fatality share is 17.1% at surface coal mines and 12.7% at surface metal and nonmetal mines, against 1.3% in underground coal and 3.0% in underground metal and nonmetal mines. All 161 surface coal records and all 92 surface metal and nonmetal records fall in MSHA's face, rib, pillar, side, or highwall category. A falling highwall section or a collapsing bank engulfs equipment and people at a scale an underground rock fall between bolts rarely does; the narratives describe loaders half buried and excavator cabs crushed by falling wall material. Low frequency and high severity is exactly the profile for which incident counts are a poor guide and early warning is most valuable.

Exposure-adjusted, surface rates are low. Divided by hours, surface rates are far below underground ones: over 2000–2025 the injury rate was 0.829 per 200,000 hours in underground coal and 0.0032 at surface metal and nonmetal mines, a ratio of about 259 to one, on 1.87 and 4.16 billion hours respectively. The fatality rate differs less, about 28 to one (5.5 against 0.2 per 100 million hours). Both views matter: per hour a surface worker is far safer from ground failure, but when a surface wall does fail, it is far more likely to kill.

Finding 2: Rates Have Fallen Unevenly, and Deaths Have Not Followed Injuries Everywhere

Injury rates fell in every setting. In underground coal the yearly injury rate fell from 1.18 in 2000 to 0.38 in 2025, and in underground metal and nonmetal mines from 0.62 to 0.13 (Figure 2). Pooled into five-year periods to damp year-to-year noise (Table 3), the underground coal rate fell 37% from 2000–04 to 2020–25, after peaking in 2005–09, and the underground metal and nonmetal rate fell 59%. Surface rates fell by about three quarters (73% for coal and 74% for metal and nonmetal) from much lower levels, on very small counts.

Fatality rates did not follow everywhere. In underground coal the fatality rate fell with the injury rate, from 7.0 to 3.1 per 100 million hours (25 and 9 deaths; 95% intervals 4.5–10.3 and 1.4–5.9). In underground metal and nonmetal mines it did not: 6.0 in 2000–04 and 5.5 in 2020–25 (5 and 7 deaths; intervals 1.9–14.0 and 2.2–11.3), while the injury rate fell by more than half. The intervals overlap, so the data do not show an increase; they show that the decline in injuries has not been matched by a measurable decline in deaths. A falling injury rate with a flat fatality rate is consistent with controls that reduce frequent small falls without reducing rare large ones, and it is the pattern that surveillance by injury counts alone would miss.

Commodity. Among underground metal and nonmetal mines, metal mines have the highest injury rate (0.391 per 200,000 hours, against 0.249 for stone and 0.210 for nonmetal), while underground stone mines have the highest fatality rate (6.5 per 100 million hours, against 5.2 for metal). Appendix D gives every commodity.

Geography. Ground-failure injuries concentrate where underground mining concentrates. WV, KY, IL account for 70% of underground coal injuries among 17 states with underground coal hours; among underground metal and nonmetal mines, the top three states account for 46% across 39 states. Rates vary as much as counts: Idaho's underground metal and nonmetal mines recorded 127 injuries at 1.36 per 200,000 hours, against 90 at 0.17 in Nevada, which worked several times as many hours. Surface metal and nonmetal injuries are spread thinly across 55 states. Appendix C lists the leading states in every setting.

Finding 3: The Federal Record Rarely Says How Surface Walls Fail

Mechanism is seldom stated. Of the 92 surface metal and nonmetal narratives (48 stone, 22 metal, 14 sand and gravel, 8 nonmetal; 10 fatal), the rules label 41 as rockfall and 21 as collapse with no mechanism stated (Figure 3, Appendix Ta-

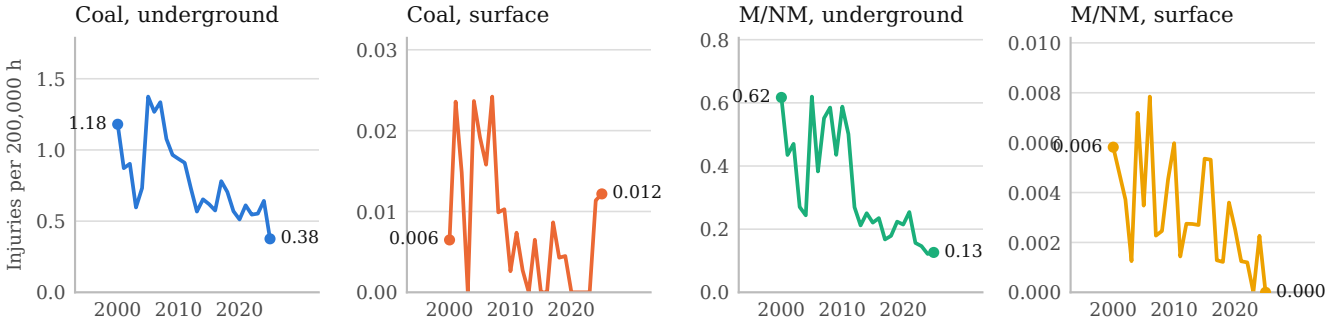


Figure 2: Ground-failure injury rate per 200,000 hours worked by mine-operator employees, by setting, 2000–2025. Each panel has its own scale; surface rates are two orders of magnitude lower than underground rates. Labels mark the first and last years.

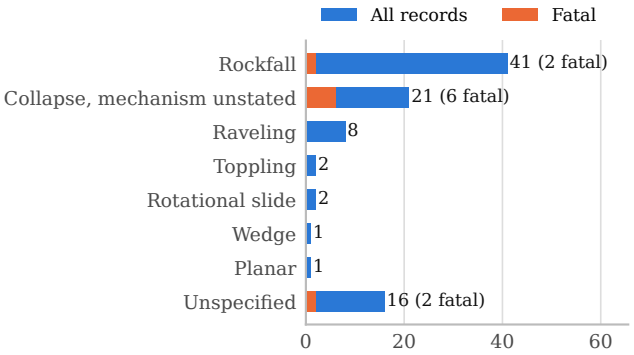


Figure 3: Rule-based failure modes of 92 surface metal and nonmetal ground-failure narratives, 2000–2026, with fatal records overlaid. Rules version 1.0.

ble 9). Only 6 narratives (7%) name a structurally controlled mechanism (wedge 1, planar 1, toppling 2, rotational slide 2), and 16 (17%) match no pattern. The collapse class carries 6 of the 10 deaths: the most lethal events are the ones whose mechanism is least often recorded. Table 4 shows narratives from each class.

When a narrative does describe the mechanism, it can be precise: one operator reported "multiple geologic faults along the highwall" acting as release planes for "a toppling deformation mechanism". Such reports show that the information exists at the mine. The reporting form simply does not ask for it.

Triggers are seldom recorded. Only 19 narratives (21%) mention any trigger: blasting 6, water 8, scaling 6. Several describe failures during or just after the work that loads or disturbs a wall: loading blast holes beneath a highwall, cleaning shot rock at the toe, scaling with a rock breaker. The excavator, loader, haul truck, drill, shovel, or dredge appears in 44 narratives (48%), often as the thing struck: equipment working at the toe of a wall is where surface ground failures meet people.

Rule errors found by inspection. Because v1.0 has no formal agreement study, we report the systematic errors we found by reading the labelled narratives (Table 5). The word "slough" sends two very large highwall failures, one of about 1.06 million tons across ten benches, to *raveling*, a label meant for progressive small-scale loss. "Slid" sends a freeze–thaw rock slide to *planar* without evidence of a sliding plane. The water trigger fires on water as a location (a wall falling into a dredge pond) as well as a cause. And MSHA's fall category itself contains at least one record that is not a ground failure: a stone ejected from a cone crusher. These errors point to specific rule changes (volume thresholds for raveling, requiring a named discontinuity for planar, distinguishing causal from locative water) that we plan for the next version, alongside a blind agreement study.

The classification misses comparable events. Screening the *other* MSHA categories at surface metal and nonmetal mines yields 226 narratives that read as ground failures, 2.5 times the 92 records selected by MSHA's fall categories, including 13 fatal records (Appendix E). They are filed mostly as powered haulage (52) and machinery (48), because the reporting form asks for the circumstance that most directly contributed, and an operator may reasonably name the loader a highwall fell on. Table 6 shows examples, including one false positive: the screen is deliberately broad and its output is a review list, not a count. Even if only a fraction are confirmed, surface ground failures are undercounted by any analysis, including ours, that relies on the classification alone.

Discussion

GroundFall-US shows a hazard that has receded in frequency but not in lethality. Underground, decades of roof bolting, screen, and rib support have cut injury rates, yet underground metal and nonmetal fatality rates have not fallen with them. At the surface, ground failures are rare per hour worked, but 14.9% of the injuries they cause are fatal, and the federal record seldom says how the wall failed or what preceded it. Counting injuries alone would understate both problems.

Recommendations. First, MSHA's Part 50 reporting could add a small set of structured fields for ground-failure events: failure mode (from a short list such as the one used here), approximate volume or dimensions, wall or bench height,

Wedge. (metal, NV, 2014; no injury) *"A wedge (block of rocks) came out of the wall along side of the haul road and struck Loader 377 while it was being used for high wall cleanup"*
Toppling. (metal, NV, 2020; no injury) *"Multiple Geologic faults along the highwall contributed to a parallel joint orientation that dips steeply to the southwest. Bounding the instability to the north & south vertical faults. These faults acted as "release" planes that resulted in a toppling deformation mechanism and movement of the highwall within this narrow sector."*
Rotational slide. (metal, MT, 2005; no injury) *"Due to high rain fall the Highwall was saturated with moisture and was being monitered. Highwall activity indicated to evacuate employees from the area. Following our safety procedures employees were evacuated. No employee was injured or eqipment was damaged. There was no rock burst just a large slump of wet material from the highwall onto 5175 bench."*
Collapse, mechanism unstated. (stone, OR, 2020; days away and restricted work) *"Excavator was producing material from pit floor in front of lower bench of highwall. For unknown reason highwall gave way & material made contact w/ excavator & cab. Material entered cab & EE received cut above lt eye at hairline on scalp. EE was taken to ER for check up & 1st Aid on Cut. On 9/21/20 EE went back for x-rays which showed 6 cracked ribs & a cracked sternum."*
Rockfall. (metal, NV, 2000; restricted work) *"WHILE BEING LOADED AT THE SHOVEL A LARGE ROCK ROLLED FROM THE HIGHWALL STRIKING THE REAR OF THE EMPLOYEE'S HAUL TRUCK. THE TRUCK ROCKED RESULTING IN MINOR NECK STRAIN. EMPLOYEE GIVEN 3 DAYS LIGHT DUTY BY PHYSICIAN."*

Table 4: Verbatim MSHA narratives by rule label (commodity, state, year; degree of injury). Narratives are public records, quoted as filed.

and observed precursors or triggers (blast, rain, freeze–thaw, scaling, undercutting). Operators already know these facts, as the most detailed narratives show; recording them in fields rather than free text would make national analysis routine. Second, any classification-based count of surface ground failures should be paired with a narrative screen like ours, because the circumstance code alone misses events. Third, the low-frequency, high-severity profile of surface failures argues for investment in monitoring (radar, prism, and visual inspection programs under 56.3130) at walls where equipment works at the toe, which is where the narratives place most victims. GroundFall-US offers a baseline against which such programs can be evaluated.

Future work. The next version will add a blind agreement study of the coding rules, expert review of the 226 candidates, the rule changes identified above, and historical rather than current mine attributes. The same pipeline extends naturally to underground narratives, where rib and roof falls could be coded by support condition.

Labelled raveling; a ten-bench highwall failure. (metal, NV, 2020; no injury) *"On 12-21-2020 at approximately 1:19pm the 5400 southwest upper highwall failed causing a slough of 10 benches and approximately 1.06 million tons. No injury or loss occurred during this incident."*
Labelled raveling, blasting; a 500,000-ton slough. (metal, NV, 2006; no injury) *"On 9/21/06 a blast pattern was detonated in the main pit and seconds later the south high wall just above the blast sloughed down. Appoximately 500,000 tons."*
Labelled planar; mechanism not stated. (stone, MT, 2006; days away from work) *"EE was injuried during a rock slide that came loose due to thawing and freezing temps. EE saw the slide and tried to move out of the way, and the slide caught his feet and threw him down and covered him to his waist. Two others moved the excavator bucket to block any more sliding and carefully moved him to the safety zone. Transported to the hospital in helicopter."*
In MSHA's fall category; not a ground failure. (sand and gravel, WI, 2010; days away from work) *"Checking water valve, stone came out of the cone crusher and struck EE on top of hard hat pushing it down causing cut on top of scalp."*

Table 5: Labelling and selection errors found by inspection (rules version 1.0).

Conclusion. We released an open, documented, reproducible dataset of 29,851 ground-failure records at U.S. mines from 2000 to 2026, with rates per hour worked and a transparent failure-mode coding layer. Ground failures cause 13.3% of mining deaths in MSHA's record; surface failures are 10 times as lethal per injury as underground ones; injury rates have fallen faster than fatality rates in underground metal and nonmetal mining; and the federal record rarely captures failure mechanism or trigger. We offer the dataset as a national reference for MSHA, NIOSH, operators, and researchers, and as the evidence base for early warning.

Data and Code Availability

Data, code, coding rules, documentation, and the three source files are at https://github.com/Lookman1234/groundfall-us (release v1.0.0), and will be archived on Zenodo with a DOI at release. Data and documentation are licensed CC BY 4.0 and code MIT. The README gives the commands to rebuild every table and figure; the source fingerprints are in Appendix F.

Acknowledgments

We thank the Mine Safety and Health Administration for maintaining its open data portal.

Highwall toppling, filed as powered haulage. (MSHA category: powered haulage; 2001; days away from work) *"EE WAS BACKING UP LOADER WHEN 80' X 54' X 4' SECTION OF HIGHWALL TOPPLED ONTL LOADER. HE WAS ENTRAPPED FOR ABOUT 1-1/2 HOURS. HE WAS EXTRICATED & TRANSPORTED TO HOSPITAL FOR EVALUATION."*
Fatal slope collapse, filed as falling materials. (MSHA category: falling/sliding/rolling materials; 2015; fatal) *"The mine slope collapsed trapping our miner in his machine. The recovery effort was not successful and the employee expired at the mine site."*
Bench failure, filed as slip or fall of person. (MSHA category: slip or fall of person; 2001; days away from work) *"RUNNING A BURNER ON A GRANITE BENCH AND THE BENCH SLID DOWN 20' TAKING EQUIPMENT & EE."*
Highwall slough, filed as machinery. (MSHA category: machinery; 2011; days away and restricted work) *"Miner was operating rock drill on 52' highwall, bit stuck & attempt to get bit unstuck,diagaonal section of the highwall sloughed,drill steel did not break,mtrl around drill steel pulled drill down with it approx. 25' & drill rolled over onto right side. Miner received laceration to forehead, laceration to right knee & contusions to right hip from wearing the seat-belt in drill."*
Not a ground failure (screen false positive). (MSHA category: slip or fall of person; 2000; days away from work) *"EMPLOYEE WAS WALKING OFF A BERM. A ROCK ROLLED UNDERFOOT CAUSING THE EE'S KNEE TO TWIST. THE EE WENT TO THE DR ON 12/18/00 AND WAS SCHEDULED FOR SURGERY. EE ALSO HAD A PRE-EXISTING CONDITION F ROM AN "OFF-THE-JOB" INCIDENT."*

Table 6: Candidate ground failures filed outside MSHA's fall categories at surface metal and nonmetal mines.

Ethics Statement

This study uses only public administrative records. MSHA publishes accident narratives, mine identifiers, and operator names as open government data; the narratives contain no worker names, and we quote them verbatim, as filed, identifying each by commodity, state, and year only. We made no contact with mines, operators, or workers. Narratives describe injuries and deaths of real people; we quote them to document the hazard and the limits of the record, not to assign fault, and nothing here is a finding about the conduct of any operator.

Use of AI assistance. Code for fetching, joining, and analysing the data, the initial draft of the coding rules, and drafting of this manuscript were assisted by an AI system (Claude, Anthropic). The authors directed the work, made the analytic decisions, approved the coding rules, and are responsible for the content. Every number in the paper is generated by the released code from the released data.

Positionality. The coding rules reflect the judgment of the first author, who wrote them. Rules written by a different expert might draw some boundaries differently, which is why every rule and the evidence phrase behind every label are released for inspection.

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Appendix A: Codebook (Incident Table)

Column	Source	Definition
document_no	Accidents	MSHA accident form number; unique key.
mine_id	Accidents	Seven-digit MSHA mine identifier.
mine_name, state, county, latitude, longitude	Mines	Current mine name and location.
accident_date, year	Accidents	Date of the event and calendar year.
sector	Derived	Coal or metal/nonmetal, from COAL_METAL_IND.
subunit_cd, subunit, mine_type	Accidents, derived	Location within the mine and grouped mine type (underground, surface, mill/plant, shop/yard, office).
commodity, primary_sic	Mines	Grouped canvass (coal, metal, nonmetal, stone, sand and gravel) and SIC description.
msha_classification	Accidents	MSHA circumstance category (selection variable).
degree_injury_cd, severity	Accidents, derived	Degree of injury and grouped severity (fatal, lost time, other reportable injury, no-injury accident, other).
is_fatal, is_injury, is_lost_time	Derived	Flags from the degree of injury.
is_contractor, contractor_id	Accidents	Contractor involvement.
occupation, activity, mining_equipment	Accidents	As reported.
has_hours_same_year	Derived	Hours exist for the same mine, subunit, and year.
narrative	Accidents	Operator narrative, as filed.

Table 7: Incident-table columns. The rate table adds hours, average employees, counts by severity, and rates; the failure-mode table adds failure_mode, trigger flags, and the evidence phrase for each label. Full definitions are in docs/CODEBOOK.md.

Appendix B: Coding Rules (Version 1.0)

Label (in precedence order)	Patterns	First pattern (regular expression) and exclusion
Wedge	2	\bwedge\w*\b; excludes wedged (between|under|in)
Planar	4	\bplan(e|ar) failure
Toppling	3	\btoppl\w*
Rotational slide	5	\b(slump\w*|rotational|circular failure)\b
Raveling	4	\bravel\w*
Rockfall	8	\b(rock|boulder|stone|piece of (rock|slate|alluvium|material)|chunk|slab)s? .{0,30}\b(fell|falling|came down|came loose|rolled|rolling|dislodged?|broke loose)
Collapse, mechanism unstated	9	\b(high ?wall|hw|pit wall|wall|bank|slope|bench|face|stockpile|spoil|overburden|over burden)s? .{0,30}\b(gave way|failed|failure|slid|slide|collaps\w*|cav(ed|ing)|subsid\w*|subsud\w*)
Unspecified	–	no mode pattern matched
Trigger: blasting	5	\bblast\w*; excludes \bshot ?crete\b
Trigger: water	3	\brain\w*; excludes \bwater truck\b
Trigger: scaling	3	\bscal(e|ed|er|ing)\b

Table 8: Coding rules. Modes are tested in the order listed and the first match wins; triggers are independent. The full rule set is config/coding_rules.yaml.

Failure mode	Records	%	Fatal
Rockfall	41	45	2
Collapse, mechanism unstated	21	23	6
Raveling	8	9	0
Toppling	2	2	0
Rotational slide	2	2	0
Wedge	1	1	0
Planar	1	1	0
Unspecified	16	17	2

Table 9: Failure-mode distribution, surface metal and nonmetal ground failures, 2000–2026 (n = 92).

Appendix C: Leading States

Setting	States	Top-3 %	Five states with most injuries: count (rate per 200,000 hours, underground only)
Coal, underground	17	70	WV 2592 (0.86); KY 2021 (1.05); IL 820 (1.16); PA 746 (0.72); VA 471 (0.77)
Coal, surface	26	62	KY 19; WV 15; VA 8; AL 7; OH 6
M/NM, underground	39	46	MT 164 (0.56); ID 127 (1.36); NV 90 (0.17); WY 67 (0.34); TN 46 (0.32)
M/NM, surface	55	23	GA 6; NV 5; PA 4; TX 4; IL 3

Table 10: Ground-failure injuries by state, mine-operator employees, 2000–2025. "States" counts states with hours worked in the setting.

Appendix D: Commodities

Commodity	Type	Hours (millions)	Injuries	Deaths	Injury rate	Fatality rate
Coal	Surface	1,513	68	11	0.009	0.7
Coal	UG	1,868	7,742	103	0.829	5.5
Metal	Surface	810	11	0	0.003	0.0
Nonmetal	Surface	351	7	2	0.004	0.6
Sand and gravel	Surface	1,435	13	2	0.002	0.1
Stone	Surface	1,562	35	4	0.004	0.3
Metal	UG	289	565	15	0.391	5.2
Nonmetal	UG	127	133	4	0.210	3.2
Stone	UG	107	133	7	0.249	6.5

Table 11: Ground-failure injuries and deaths by commodity and mine type, mine-operator employees, 2000–2025. Injury rate per 200,000 hours; fatality rate per 100 million hours. Settings with under one million hours are omitted.

Appendix E: Supplement Candidates by MSHA Category

MSHA classification	Candidates	Fatal
Powered haulage	52	3
Machinery	48	3
Slip or fall of person	40	0
Handling of materials	34	0
Falling/sliding/rolling materials	16	5
Handtools (nonpowered)	12	0
Other	6	2
All other categories	18	0

Table 12: Surface metal and nonmetal narratives outside MSHA's fall categories flagged by the screen (n = 226), 2000–2026. Candidates await expert review and are not in the v1.0 counts.

Appendix F: Reproducibility

The pipeline runs in eight numbered scripts (fetch, build, QA, coding, candidate screen, validation sample, figures, summary) plus the script that generates every number in this paper. Run with the archived source files, it reproduces every table byte for byte; we verified this with an independent rebuild. Source files and fingerprints:

File	Bytes	SHA-256
`Accidents.zip`	52,269,752	`62d0c861a5c3a8e76bd06de79ac60e4d` `cc643bca148501cb3fc752fa76c6687d`
`Mines.zip`	7,308,890	`1a0957ff00496855289725e351797d17` `37faad53c48ef7bf5ac51d725e997e02`
`MinesProdYearly.zip`	7,072,831	`c0fdca86d1756fc03b3f22dbc9430a02` `cefae9a8afac5d8e9e8503f590a3c342`

Table 13: MSHA source files, edition of 9 October 2026.